

**Instructions:**

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

**Mathematics**

1. In a triangle, the measure of one angle is  $48^\circ$ . The second angle is twice the third angle. **What is the measure of the largest angle of the triangle?**  
(A)  $48^\circ$  (B)  $66^\circ$   
(C)  $88^\circ$  (D)  $96^\circ$
2. In a school exhibition, the ratio of handmade items to total items displayed was 7:12. If 180 items were displayed in total, **how many handmade items were there?**  
(A) 105 (B) 108  
(C) 112 (D) 120
3. A farmer wants to fence a square field and also divide it into 4 equal square plots using fencing inside. If the side of the field is 20 m, **what is the total length of fencing required?**  
(A) 160 m (B) 120 m  
(C) 200 m (D) 240 m
4. **Statement–1:** The product of  $3x^2$  and  $5x^3$  is  $18x^5$ .  
**Statement–2:** While multiplying powers with the same base, their exponents are added.  
(A) Both Statement I and Statement II are true  
(B) Both Statement I and Statement II are false  
(C) Statement I is true and Statement II is false  
(D) Statement I is false and Statement II is true
5. A shopkeeper mixes rice worth ₹48/kg and ₹60/kg in the ratio 5:3. **What is the cost per kg of the mixture?**  
(A) ₹52.5  
(B) ₹54  
(C) ₹55.5  
(D) ₹56

6. A school records the number of students absent for 6 days as:  
12, 15, 9, 18, 6, 15.

The school plans an awareness drive on days when absentees are **above the average**. On **how many days will the drive be conducted?**

- (A) 2 (B) 3  
(C) 4 (D) 5

7. A stack of 12 identical math books is 33.6 cm high. **What is the thickness of one book in millimeters?**

- (A) 2.8 mm  
(B) 28 mm  
(C) 280 mm  
(D) 0.28 mm

8. Which of the following sets of lengths can form a **triangle**?

- (A) 4 cm, 5 cm, 9 cm  
(B) 6 cm, 7 cm, 10 cm  
(C) 3 cm, 4 cm, 7 cm  
(D) 8 cm, 2 cm, 10 cm

9. A water tank can be filled by 12 pipes in 10 hours. **How long will 15 pipes take to fill the same tank?**

- (A) 6 hours  
(B) 7 hours  
(C) 8 hours  
(D) 9 hours

10. A classroom floor is 8 m long and 6 m wide. A carpet of width 1 m is laid along the inner boundary of the floor. **Find the area left uncovered.**

- (A)  $12 \text{ m}^2$  (B)  $16 \text{ m}^2$   
(C)  $20 \text{ m}^2$  (D)  $24 \text{ m}^2$

11. The diagonals of a rhombus are in the ratio 3:4. If its area is  $96 \text{ cm}^2$ , **find the length of the larger diagonal.**

- (A) 12 cm (B) 16 cm  
(C) 20 cm (D) 24 cm

12. A shopkeeper sells the same type of notebook to students. The cost of one notebook is ₹x.

After selling 8 notebooks, the total money collected is ₹480, which is ₹120 more than twice the cost of one notebook.

**Which of the following equations correctly represents this situation?**

- (A)  $8x = 2x + 120$   
(B)  $8x - 120 = 2x$   
(C)  $2x - 120 = 8x$   
(D)  $8x + 120 = 2x$

13. If the perimeter of an equilateral triangle is  $3(2x + 1)$  cm, find the length of one side when  $x = 3$ .

- (A) 5 cm (B) 7 cm  
(C) 9 cm (D) 11 cm

14. Match the given dimensions with the correct area.

| Column I                                           | Column II             |
|----------------------------------------------------|-----------------------|
| (i) Square of side 9 cm                            | (P) $40 \text{ cm}^2$ |
| (ii) Rectangle of length 8 cm and breadth 5 cm     | (Q) $42 \text{ cm}^2$ |
| (iii) Triangle with base 12 cm and height 7 cm     | (R) $81 \text{ cm}^2$ |
| (iv) Parallelogram with base 14 cm and height 6 cm | (S) $84 \text{ cm}^2$ |

- (A) i-(P), ii-(Q), iii-(R) iv-(S)  
 (B) i-(P), ii-(R), iii-(Q), iv-(S)  
 (C) i-(R), ii-(P), iii-(Q), iv-(S)  
 (D) i-(S), ii-(R), iii-(Q), iv-(P)

15. Rahul reads  $\frac{1}{5}$  of a book on Monday. On Tuesday, he reads  $\frac{3}{4}$  of the remaining pages. If 40 pages are still left to be read, **what is the total number of pages in the book?**

- (A) 200 (B) 160  
 (C) 100 (D) 240

16. The population of bacteria doubles every hour. If the initial population is  $2^4$ , **what will be the population after 6 hours?**

- (A)  $2^{10}$  (B)  $2^{24}$   
 (C)  $2^4$  (D)  $2^6$

17. Find the value of:  $(2^3)^4 \div 2^5$

- (A)  $2^7$  (B)  $2^{17}$   
 (C)  $2^1$  (D)  $2^{12}$

18. A pictograph uses the scale  $\star = 20$  notebooks. If a shopkeeper sold  $\star \star \star \frac{1}{2}$  notebooks in a day, **how many notebooks were sold?**

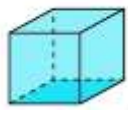
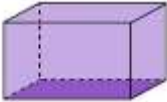
- (A) 60 (B) 70  
 (C) 75 (D) 80

19. A man spends 25% of his income on rent and 40% of the remaining on food. If he saves ₹4500, find his income.

- (A) ₹10,000 (B) ₹12,000  
 (C) ₹15,000 (D) ₹18,000

20. A solid has 6 rectangular faces, 12 edges, and 8 vertices.

If its top view is a rectangle and all vertical edges are equal, the solid is a:

- (A)  (B)   
 (C)  (D) 

21. A car travels 180 km at a constant speed. If the speed was 15 km/h more, it would take 2 hours less. **What was the original speed of the car?**

- (A) 30 km/h (B) 45 km/h  
 (C) 60 km/h (D) 75 km/h

22. A triangular banner has a base of 20 cm.

Its height is **3 cm less than half of the base**. What is the **area of the banner**?

- (A) 70 sq. cm                      (B) 80 sq. cm  
(C) 85 sq. cm                      (D) 90 sq. cm

23. In a school, the ratio of boys to girls is 5:4. If 20% of the boys and 10% of the girls are absent on a certain day, **what is the ratio of boys present to girls present**?

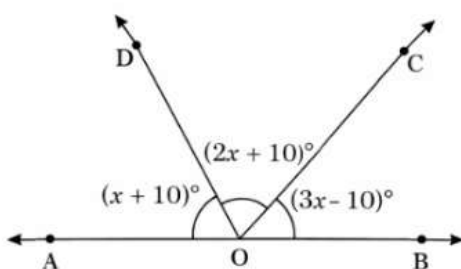
- (A) 10:9                              (B) 4:3  
(C) 5:4                                (D) 6:5

24. The area of a square is  $16 \times 10^4 \text{ cm}^2$ .

**Find the length of its side.**

- (A)  $4 \times 10^2 \text{ cm}$                       (B)  $2 \times 10^2 \text{ cm}$   
(C)  $8 \times 10^4 \text{ cm}$                       (D)  $16 \times 10^2 \text{ cm}$

25. In the given figure **find the value of x**



- (A)  $30\frac{1}{3}^\circ$                                   (B)  $40^\circ$   
(C)  $28\frac{1}{3}^\circ$                                   (D)  $50^\circ$

26. Three angles on a straight line are in the ratio **1 : 2 : 3**. **What is the measure of the largest angle?**

- (A)  $30^\circ$                                   (B)  $60^\circ$   
(C)  $90^\circ$                                   (D)  $120^\circ$

27. A rectangular park has a length of  $24\frac{1}{2}$  meters and a breadth of 18.4 meters. If a boy runs around the park 4 times, **what is the total distance he covers?**

- (A) 343.2 m  
(B) 171.6 m  
(C) 328.4 m  
(D) 85.8 m

28. A square courtyard and a rectangular courtyard have the same perimeter. The side of the square is 18 m. If the rectangle's length is 30 m, **what is its breadth?**

- (A) 6 m                                  (B) 9 m  
(C) 12 m                                (D) 15 m

29. Two triangles are congruent by SSS. One triangle has sides 3 cm, 5 cm, and  $(x + 2)$  cm. The corresponding sides of the other triangle are 3 cm, 5 cm, and 9 cm. **Find x.**

- (A) 5                                      (B) 6  
(C) 7                                      (D) 8

30. A shopkeeper sells 3 identical pens and 2 notebooks for ₹94. The same pens and notebooks together, but with 1 extra notebook, cost ₹124. **What is the cost of one notebook?**

- (A) ₹10                                  (B) ₹15  
(C) ₹30                                  (D) ₹25

31. If the ratio of milk to water in a mixture is 7:3 and 20 litres of water is added, the ratio becomes 7:5. **Find the original quantity of the mixture.**

- (A) 50 litres (B) 60 litres  
(C) 100 litres (D) 80 litres

32. A rope is cut into **three pieces** of lengths  $(2x+1)$  m,  $(x-3)$  m, and  $(3x+2)$  m. Which expression represents the **total length** of the rope?

- (A)  $6x+1$  (B)  $6x+6$   
(C)  $6x$  (D)  $5x$

33. Simplify:  $\frac{(3^2 \times 10^3)^2}{3^2 \times 10^4}$

- (A)  $9 \times 10^2$  (B)  $3 \times 10^4$   
(C)  $9 \times 10^4$  (D)  $3 \times 10^2$

34. Two right-angled triangular ramps are constructed for wheelchair access. Each ramp has a hypotenuse of 13 m and one side of 5 m.

**Which condition ensures both ramps are exactly the same in shape and size?**

- (A) SSS (B) ASA  
(C) SAS (D) RHS

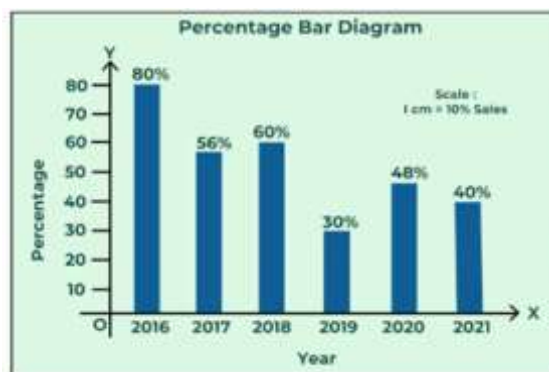
35. The temperature in a hill station in the morning was  $-3^\circ\text{C}$ . By afternoon, the temperature rose by  $5^\circ\text{C}$ . **What was the temperature in the afternoon?**

- (A)  $-8^\circ\text{C}$  (B)  $-2^\circ\text{C}$   
(C)  $2^\circ\text{C}$  (D)  $8^\circ\text{C}$

36. Two angles are formed on a straight road when a side lane joins it. If one angle is  $(3x + 15)^\circ$  and the other is  $(5x - 15)^\circ$ , **what is the value of the larger angle?**

- (A)  $75^\circ$  (B)  $97.5^\circ$   
(C)  $105^\circ$  (D)  $120.5^\circ$

37. Read the bar graph carefully and answer the following question.



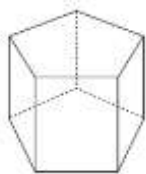
If the sales percentage in 2022 increases by **25% of the 2021 value**, then the sales percentage in **2022** will be:

- (A) 45% (B) 48%  
(C) 50% (D) 52%

38. During a school exhibition, a ribbon of length  $\frac{9}{4}$  **metres** is used to decorate notice boards. The ribbon is cut into pieces of equal length  $\frac{3}{8}$  **metres** each. **How many such pieces can be obtained?**

- (A) 4 (B) 5  
(C) 6 (D) 8

39. Which of the following **statements** is **CORRECT** for the given solid?



- (A) It has 5 faces, 8 edges and 5 vertices
- (B) It has 7 faces, 10 edges and 10 vertices
- (C) It has 7 faces, 15 edges and 10 vertices
- (D) It has 10 faces, 15 edges and 7 vertices
40. Five years ago, the age of a father was three times the age of his son. After 5 years, the father's age will be twice the son's age. **What is the present age of the son?**
- (A) 10 years                      (B) 12 years
- (C) 15 years                      (D) 20 years
41. A climber climbs 120 m, then slips down 185 m, and finally climbs 60 m. **What is his final position with respect to the starting point?**
- (A) -5 m                          (B) -65 m
- (C) -125 m                      (D) -245 m
42. What number must be added to **8, 18, and 38** so that they are in **continued proportion**?
- (A) 2                              (B) 4
- (C) 6                              (D) 8

43. Consider a complex simplification involving mixed fractions.

If we subtract the sum of  $3\frac{5}{12}$  and  $2\frac{1}{3}$  from the sum of  $5\frac{5}{6}$  and  $4\frac{1}{4}$ , **what is the final result?**

- (A)  $4\frac{1}{4}$                               (B)  $3\frac{3}{4}$
- (C)  $4\frac{1}{3}$                               (D)  $3\frac{11}{12}$
44. The profit of a shopkeeper one day was ₹350. The next day there was no profit and no loss. **What integer represents the total change?**
- (A) 350                              (B) -350
- (C) 0                                  (D) 700
45. A cyclist covers a distance of 60 km. If his speed is increased by 5 km/h, he takes 1 hour less to cover the same distance. **What was his original speed?**
- (A) 10 km/h
- (B) 12 km/h
- (C) 15 km/h
- (D) 20 km/h
46. The perimeter of an equilateral triangle is numerically equal to the perimeter of a square. If the side of the square is 12 cm, **what is the side of the triangle?**
- (A) 12 cm                              (B) 14 cm
- (C) 16 cm                              (D) 18 cm

47. The ratio of boys to girls in a class is 5:3.

If 10 more girls join, the ratio becomes 5:4. **Find the original number of students.**

(A) 64 (B) 72

(C) 80 (D) 96

48. A shopkeeper loses ₹75 every day for 6 days. **What integer represents his total loss?**

(A) 450 (B) -450

(C) -81 (D) 81

49. Two straight roads intersect at a point forming four angles. One of the angles

measures  $38^\circ$ . A traffic engineer wants to place signal boards at the angle directly opposite to it. **What will be the measure of the angle where the board is placed?**

(A)  $38^\circ$  (B)  $52^\circ$

(C)  $142^\circ$  (D)  $180^\circ$

50. The total cost of 4 kg of apples and 3 kg of oranges is ₹260. If the cost of 1 kg of apples is ₹20 more than 1 kg of oranges, **what is the cost of 1 kg of oranges?**

(A) ₹25.71 (B) ₹30

(C) ₹40.5 (D) ₹50

